

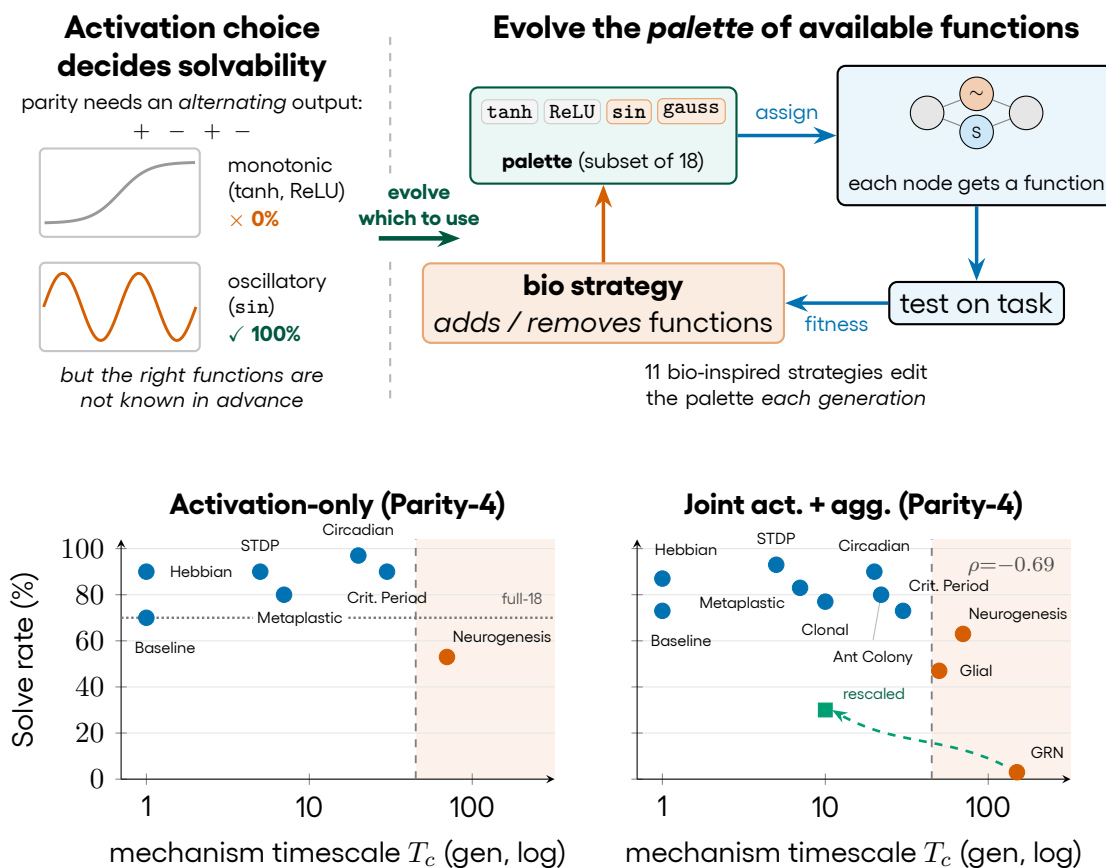
Don't hand-pick per-node activation functions.

Let evolution learn which to use,
with bio-inspired meta-learning strategies.

Bio-Inspired Palette Evolution in Indirectly Encoded Substrates: Timescale Compatibility Shapes Activation Function Discovery

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Solve rate falls as the adaptation cycle T_c grows (not its solving speed). Both panels single-task Parity-4: activation-only (left, 7 strategies); the joint act. + agg. system (right, 11) adds slow mechanisms (Clonal, Ant Colony, Glial, GRN) that drive the T_c -solve-rate correlation ($\rho = -0.69$). Dotted line: the static full-18 palette (no strategy; 70% left), a non-timescale control.

The problem

Indirect substrate encoding lets every node pick its activation from a *palette*, but which functions belong in it? Monotonic-only palettes leave parity **unsolvable**; the full 18-function set **underperforms** a curated one.

The approach

Treat palette composition as **meta-learning**: 13 strategies (11 bio-inspired) edit the palette from population fitness each generation. 30–60 replications. Activation alone, then a *second* aggregation palette evolved the same way.

Findings

Same solve rate, ~2× faster: vs a tuned baseline; the Circadian Strategy halves total compute.

Timescale shapes success: mechanisms matching the ~20-gen window win; the slowest recovers when sped up (3% → 30%).

No universal winner: the Circadian Strategy leads on parity (97%) but drops to 77% on Concentric Circles (Hebbian 90% → 53%).

TAKE-HOME

Match the mechanism's timescale to the evaluation budget.

The timescale link ($\rho = -0.69$) and ~2× speed gain are significant; gaps among the top strategies are not.

